American Terawatt

TEMPLATE TEST

Cable Link Single-Line Diagram

TikZ and circuitikz renderings of the sending-end / receiving-end transmission model

Test document for the shared report template, combining the two earlier test folders. It compiles both renderings of the single-line diagram of a cable transmission link (source, disconnectors, cable of length L_{cable} and load impedance Z_L) with the sending-end and receiving-end voltages, currents and power flows, and checks them against the original drawing.

Revision history

REV.	DATE	AUTHOR	DESCRIPTION
0.1	Sep. 18, 2026	Eugen Starschich	Initial document
0.2	Sep. 18, 2026	Eugen Starschich	`documentation/test` and `presentation/test` combined in `test/`; circuitikz variant added

1 Comparison with the original

Figure 1 reproduces the original drawing (`figures/image9.pdf`) that the TikZ figure was traced from, for a side-by-side check of geometry, colours and labels. Both figures are scaled to the full text width, so they can be compared directly.

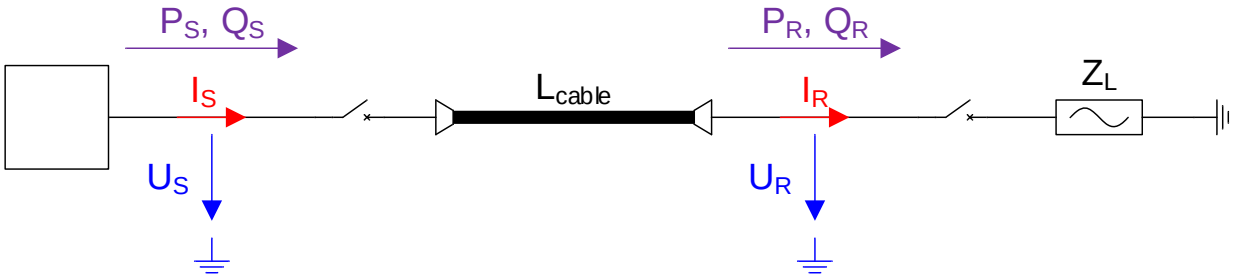


Figure 1: Original drawing used as reference.

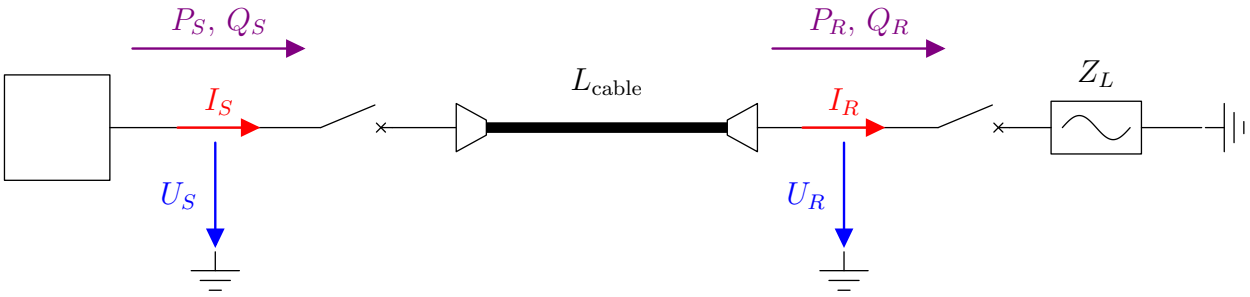


Figure 2: TikZ rendering of Figure 1 with Fable.

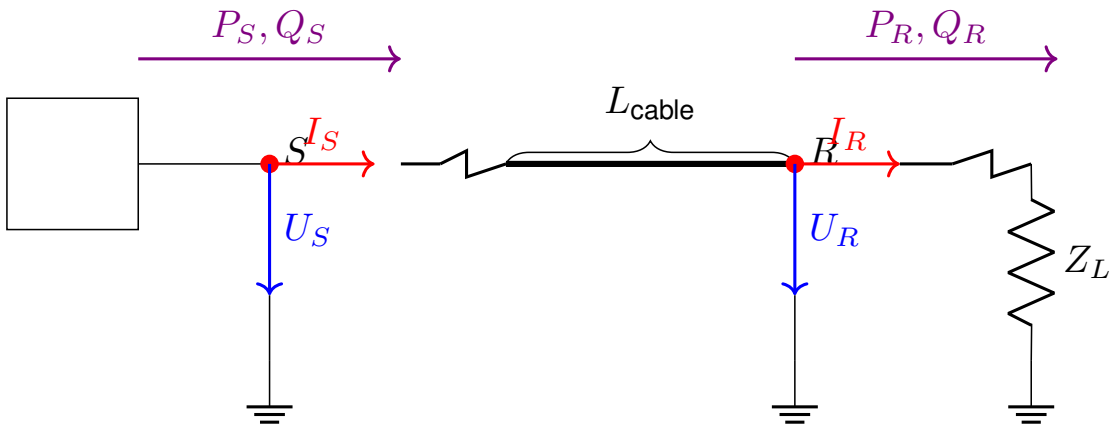


Figure 3: TikZ rendering of Figure 1 with qwen3.5:122b-a10b.

The two renderings differ in what they delegate to a package rather than in what they show.

AMERICA'S PRIVATE INDUSTRIAL POWER GRID